CHAPTER

15

X-Chromosome Analysis

The X-chromosome (ChrX) has potential forensic and human identity testing applications due to its inheritance pattern compared to other genetic markers (Szibor et al. 2003, Table 15.1). Normal males possess one X-chromosome and one Y-chromosome while females possess two X-chromosomes although there are occasionally some irregular karyotypes, such as XXY (Klinefelter syndrome; Giltay & Maiburg 2010), XXX, and XYY. More than 40 STR markers have been characterized from the X-chromosome, and population studies have been performed with many of these X-chromosome STRs (X-STRs). Literature on the use of X-chromosome analysis is growing and is briefly reviewed in this chapter. The reading list and internet resources at the back of the chapter provide over 145 references to additional information so that interested readers can dig deeper into ChrX population studies and applications.

TABLE 15.1 Specific Relationships and the Probability of Transmitting Genetic Information (Barring Mutation). Some of the ChrY Information is Not Applicable (N/A) as Women Do Not Have a Y-Chromosome.

Inheritance	Autosomal Markers	ChrY Markers	mtDNA	ChrX Markers
Mother → Son	50%	N/A	100%	100%
Mother → Daughter	50%	N/A	100%	50%
Father → Son	50%	100%	0%	0%
Father → Daughter	50%	0%	0%	100%
Paternal Grandmother → Granddaughter	25%	N/A	0%	100%
Maternal Grandmother → Granddaughter	25%	N/A	100%	25%
Paternal Grandfather → Grandson	25%	100%	0%	0%

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ChrX INHERITANCE AND APPLICATION

X-Chromosome Structure and Function

The X-chromosome was originally named X for "unknown" as early geneticists were puzzled how it remained apart from other chromosomal pairs (Gunter 2005). (Apparently when its smaller counterpart sex-chromosome was finally discovered, it received the next letter in the alphabet—Y.) With the completion of the Human Genome Project, a full sequence of human ChrX has been released and includes more than 99% coverage of the euchromatic sequence (Ross et al. 2005). The X-chromosome is 153 Mb in length and includes almost 1100 genes. ChrX represents almost 5% of the total genetic material in females but only about 2.5% in males, which only possess a single copy.

In males, the tips of the X-chromosome combine with the pseudoautosomal regions of the Y-chromosome to maintain proper segregation in cell division. In addition, there is a great deal of homology (similar or the same DNA sequence) between portions of ChrX and ChrY.

In the early embryonic development of females, one of their two X-chromosomes is inactivated or "silenced" to ensure that they only have one functional ChrX copy in each somatic cell. Either the paternal or maternal ChrX can become the active one. The inactivated X-chromosome is referred to as a Barr body after its 1949 discoverer Murray Barr (Barr & Bertram 1949). Recent research suggests that the Barr body may be more biologically active than was previously supposed (Carrel & Willard 2005). While genes are inactivated and not typically expressed from the Barr body, DNA testing results still reflect the genetic information present in both female X-chromosomes. In addition, the two female X-chromosomes can recombine during meiosis.

APPLICATIONS OF ChrX TESTING

X-chromosome STR typing can be helpful in some kinship analysis situations particularly with deficient paternity cases where a DNA sample from one of the parents is not available for testing. For example, if a father/daughter parentage relationship is in question, X-STRs may be helpful due to the 100% transmission of the father's X-chromosome to his daughter (Table 15.1). On the other hand, in a father/son parentage question, Y-chromosome results would be helpful (see Chapter 13). Table 15.2 lists several applications for X-chromosome DNA testing. ChrX testing can be especially helpful in some missing persons or disaster victim identification situations (see Chapter 9) where direct reference samples are not available and biological relatives must be sought to aid human identification.

TABLE 15.2 Applications of X-Chromosome Analysis (See Figure 15.1 for Illustration of Example Pedigrees).

- Complex kinship cases involving at least one female
- Disputed paternity to a daughter (especially in motherless cases)
- Half-sister testing where the father is the common relative
- Grandparent—grandchild comparisons
- Paternity testing in incest cases (see Figure 15.2)

X-chromosome markers can help infer parent-offspring relationships that involve at least one female, such as mother-daughter, mother-son, and father-daughter duos (illustrated in Figure 15.1). In complicated kinship scenarios, such as incest (Figure 15.2), ChrX markers may aid sorting out difficult relationship questions.

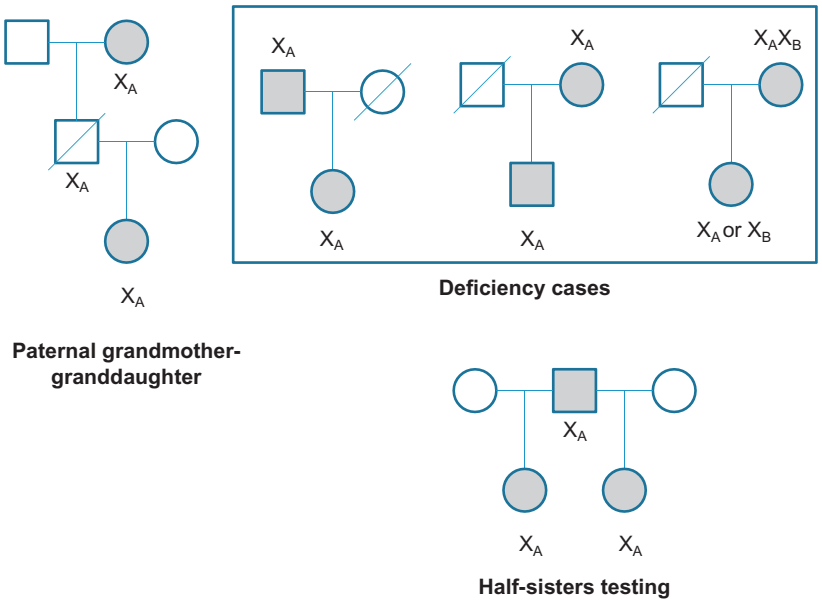


FIGURE 15.1 Some example pedigrees where ChrX testing can be helpful.

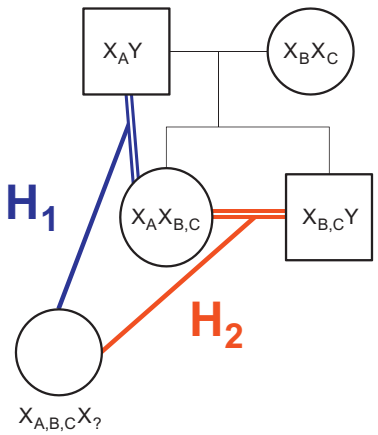


FIGURE 15.2 Use of ChrX testing in an incest case to help distinguish whether the victim's father (H_1) or brother (H_2) fathered the victim's daughter. The mother passes a combination of her X-chromosomes (X_B, X_C) on to her son ($X_{B,C}$). If either X_B or X_C is more abundant in the victim's daughter, then H_2 is more likely (her brother is the father). If X_A is more abundant in the victim's daughter, then H_1 is more likely (her father is the father). Autosomal genetic markers would probably not be very helpful in this situation due to the high degree of allele sharing expected among close relatives.

X-STR MARKERS COMMONLY USED

As with the rest of the human genome, STR markers are prevalent along the X-chromosome with comparable density to autosomal STRs (Subramanian et al. 2003). However, an enrichment of $[GATA]_n$ tetranucleotide repeats has been shown to occur in a 10 Mb portion of the short arm of ChrX (McNeil et al. 2006). Over the past decade, a number of X-STRs have been studied.

When autosomal STRs were being characterized in the early 1990s by Tom Caskey's group at Baylor, two X-STRs named HPRTB and ARA were among them (Edwards et al. 1992). However, it would be almost another decade before much was attempted with these loci in the context of X-STR typing (Szibor et al. 2003a).

Reinhard Szibor at the Institute for Legal Medicine in Magdeburg, Germany and his colleagues including Jeanett Edelmann (Leipzig, Germany) and Sandra Hering (Dresden, Germany) brought X-STR typing into use within the forensic DNA community (Szibor 2007). Professors Szibor, Edelmann, and Hering have formed a forensic ChrX research website (http://www.chrx-str.org/) with information on X-STR markers. For his efforts in this area, in 2007 Dr. Szibor was awarded the prestigious biannual Scientific Prize by the International Society of Forensic Genetics.

Table 15.3 describes 33 X-STR loci that have been used within the forensic community. As with autosomal STR loci used in forensic analysis, tetranucleotide repeats are most commonly selected due to lower stutter product formation compared to dinucleotide or trinucleotide repeats (see Chapter 5). In Table 15.3, the chromosomal position for each locus is noted as is its physical location and its genetic distance based on a linkage/recombination map created at Rutgers University (Matise et al. 2007).

TABLE 15.3 Characteristics of 33 X-Chromosome STR Loci.

Position	ChrX Marker	Repeat Motif	Allele Range	Location (Mb)	Genetic Distance (cM)	Kit/Assay (see Table 15.4)
p22.33	**DXS6807**	GATA	11–17	4.753	14.76	F
p22.32	**DXS9895**	AGAT	11–18	7.387	17.09	G
p22.31	**DXS10148**	AAGA	13.3–38.1	9.198	19.84	A
p22.31	**DXS10135**	GAAA	13–39.2	9.266	20.03	B
p22.31	**DXS8378**	CTAT	7–17	9.330	20.21	A,B,C,D,E,F,H2
p22.2	**DXS9902**	GATA	7–16	15.234	32.32	E,H1,H2
p22.11	**DXS6795**	ATT/ATC	9–16	23.254	44.24	H2
Centromere	**DXS7132**	TCTA	6–20	64.572	90.75	A,B,C,D,E,F,G,H2
q12	**DXS10079**	AGAR	14–25	66.633	90.82	A,B
q12	**HumARA**	CAG	14–31	66.682	90.81	*No longer used!*
q12	**DXS10074**	AAGA	4–21	66.894	90.83	A,B

(Continued)

TABLE 15.3 Characteristics of 33 X-Chromosome STR Loci. (*Continued*)

Position	ChrX Marker	Repeat Motif	Allele Range	Location (Mb)	Genetic Distance (cM)	Kit/Assay (see Table 15.4)
q13.1	**DXS981(STRX1)**	TATC	9.3–17	68.114	92.81	–
q13.3	**DXS6800**	TAGA	15–23	78.567	97.49	G
q21.2	**DXS6803**	TCTA	8–15	86.318	99.40	H2
q21.31	**DXS9898**	TATC	7–16	87.683	101.29	D,E,G
q21.33	**DXS6809**	ATCT	27–40	94.825	108.12	D,E,F
q21.33	**DXS6789**	TATS	10–26	95.336	108.47	D,E,F,G,H1
q22.1	**DXS7424**	TAA	7–20	100.505	115.25	F,H1
q22.1	**DXS101**	CTT/ATT	14–32	101.300	116.15	D,F,H1
q22.3	**DXS7133**	ATAG	6–14	108.928	118.18	E,F,G
q23	**GATA172D05**	TAGA	5–17	113.061	124.36	D,E,F,H2
q24	**DXS7130**	TATC	10–18.3	118.084	130.28	G,H1
q25	**GATA165B12**	AGAT	8–13	120.706	136.18	H1
q26.2	**DXS10103**	YAGA	15–21	133.246	149.37	A
q26.2	**HPRTB**	AGAT	6–19	133.443	149.66	A,B,C,D,F,G,H2
q26.3	**DXS10101**	AAAG	24–38	133.482	149.75	A,B
q27.1	**GATA31E08**	AGGG/AGAT	7–16	140.062	160.54	E,F,G,H1
q28	**DXS8377**	AGA	33–60	149.317	183.66	D
q28	**DXS10146**	TTCC/CTTT	24–46.2	149.335	183.72	A
q28	**DXS10134**	GAAA	28–46.1	149.401	183.96	A,B
q28	**DXS10147**	AAAC	6–11	149.414	184.01	H2
q28	**DXS7423**	TCCA	8–19	149.462	184.19	A,B,C,D,E,F,G,H2
q28	**DXS10011**	GRAA	18–50	150.939	188.70	G

Adapted from Machado & Medina-Acosta (2009) and http://www.chrx-str.org. Genetic distance from Rutgers map v.2 of the human genome (http://compgen.rutgers.edu/old/map-interpolator/; see Matise et al. 2007). The four linkage groups are gray-shaded. Kit/assay configurations are shown in Table 15.4. Argus X12 encompasses all Argus X8 markers. The X-STR locus HumARA is shown in red font because it is associated with disease and no longer used (Szibor et al. 2005).

When loci are physically close together on a chromosome, they are considered linked and thus not inherited independently of one another. Genetic linkage is a function of the physical distance of DNA sequences along a chromosome. As discussed with Y-chromosome markers (see Chapter 13), information from linked loci are typically treated as a haplotype rather than using the product rule to multiply allele frequencies together.

Linkage Groups and Haplotype Blocks

Four linkage groups are typically associated with the X-chromosome (see Table 15.3). When information from loci in the same linkage group is included in a set of data, it needs to be combined into a haplotype. Within a linkage group, the haplotype frequencies rather than allele frequencies are collected during population studies. With closely linked loci, the product rule cannot be used to estimate the rarity of a DNA profile by multiplying results between loci within a linkage group. However, information between linkage groups is considered independent and can be multiplied together.

Thus, alleles observed in a sample with the three X-STRs located in linkage group 1 (DXS10148, DXS10135, and DXS8378) would be combined to form a haplotype DXS10148-DXS10135-DXS8378. Frequency information from this haplotype could then be combined (multiplied) with haplotype frequency information from linkage group 2 loci (DXS7132, DXS10079, and DXS10074) to estimate the rarity of the sample profile. Some statistical formulas for use with ChrX analysis are described in D.N.A. Box 15.1 (Krawczak 2007).

X-STR LOCUS HUMARA NO LONGER USED

HumARA was an early adopted X-STR marker (Desmarais et al. 1998) and was used for a number of years until it was directly linked to genetic disease and ethical concerns led to it being dropped from active use (Szibor et al. 2005). The HumARA trinucleotide repeat is found in exon 1 of the androgen receptor gene. The CAG repeat at this locus directly codes for a polyglutamine tract. Mutations at HumARA can result in spinal and bulbar muscular atrophy when the trinucleotide repeat expands (Szibor et al. 2005).

X-STR KITS AND MULTIPLEX ASSAYS

Several X-STR assays have been developed and reported in the literature (Table 15.4). A number of population studies have been performed using these various assays.

The first commercial X-STR kit came from Biotype (Dresden, Germany) and was called the Argus X-UL (UL for unlinked). This kit contains one X-STR from each of the four linkage groups. A few years later Biotype released the Argus X-8 that added one additional X-STR to each of the four linkage groups. In 2010, QIAGEN (Hilden, Germany) partnered with Biotype and began marketing the Investigator Argus X-12 with three X-STRs in each of the four linkage groups. Figure 15.3 displays the dye-label and X-STR locus size ranges for the Argus X-12 kit.

Other Markers

As with autosomal and Y-chromosome systems, single nucleotide polymorphisms (SNPs) and insertion-deletion (INDELs) exist on the X-chromosome and assays can be designed to utilize them. A 25plex X-SNP assay has been developed and shown to have forensic

D.N.A. BOX 15.1

STATISTICAL TREATMENT OF RESULTS FROM ChrX MARKERS

Using autosomal markers, the mean exclusion chance (MEC) for an unrelated male when paternity of a daughter is disputed may be calculated using (Szibor et al. 2003).

$$\sum_i p_i^3(1-p_i)^2 + \sum_i p_i(1-p_i)^3 + \sum_{i<j} p_i p_j (p_i + p_j)(1-p_i-p_j)^2$$

where p_i refers to the frequency of the ith allele for the genetic marker being evaluated. Alternatively with X-chromosome markers, the MEC equation for paternity trios involving daughters is.

$$\sum_i p_i^3(1-p_i) + \sum_i p_i(1-p_i)^2 + \sum_{i<j} p_i p_j (p_i + p_j)(1-p_i-p_j)$$

Krawczak (2007) notes that the two formulas differ by the exponent of the last factor in each summation. With the factor always smaller in the ChrX equation, the MEC of an X-chromosome marker will be larger than an autosomal marker possessing the same allele frequencies.

Further statistical calculations involving ChrX results include (Szibor et al. 2003):

MEC for ChrX markers in trios involving daughters:

$$1 - \sum_i p_i^2 + \sum_i p_i^4 - \left(\sum_{i<j} p_i^2\right)^2$$

MEC for ChrX markers in father/daughter duos:

$$1 - 2\sum_i p_i^2 + \sum_i p_i^3$$

Power of discrimination (PD) in females:

$$1 - 2\left(\sum_i p_i^2\right)^2 + \sum_i p_i^4$$

PD in males:

$$1 - \sum_i p_i^2$$

Sources:

Krawczak, M. (2007). Kinship testing with X-chromosomal markers: mathematical and statistical issues. Forensic Science International: Genetics, *1, 111–114.*

Szibor, R., et al. (2003). Use of X-linked markers for forensic purposes. International Journal of Legal Medicine, *117, 67–74.*

usefulness (Tomas et al. 2010). Results from a set of 33 X-linked insertion/deletion polymorphisms have also been reported (Freitas et al. 2010).

POPULATION DATA COLLECTION AND OTHER STUDIES

The reference list at the end of this chapter includes 66 published population studies using a variety of X-STR markers. Data has been collected on African, Amerindian, Bosnian,

TABLE 15.4 Configuration of Several X-STR Assay and Kits Reported in the Literature Showing Dye Color Labels and Relative Sizes of PCR Products. The Arbitrary Assay/Kit Designation (A, B, C, etc.) Is Used in Table 15.3.

A	**ARGUS X-12 KIT (QIAGEN/BIOTYPE)**					
	Blue:	Amel	DXS10103	DXS8378	DXS7132	DXS10134
	Green:	DXS10074	DXS10101	DXS10135		
	Yellow:	DXS7423	DXS10146	DXS10079		
	Red:	HPRTB	DXS10148			
B	**ARGUS X-8 KIT (BIOTYPE)**					
	Blue:	Amel	DXS8378	DXS7132	DXS10134	
	Green:	DXS10074	DXS10101	DXS10135		
	Yellow:	DXS7423	DXS10079	HPRTB		
C	**ARGUS X-UL KIT (BIOTYPE)**					
	Blue:	DXS7132	DXS7423	DXS8378	HPRTB	
	Green:	Amel				
D	**PORTUGUESE 10PLEX ASSAY A (GOMES et al. 2007)**					
	Blue:	DXS8378	DXS9898	DXS8377	HPRTB	
	Green:	GATA172D05		DXS7423	DXS6809	
	Yellow:	DXS7132	DXS101	DXS6789		
E	**PORTUGUESE 10PLEX ASSAY B (GUSMÃO et al. 2009)**					
	Blue:	DXS8378	DXS9898	DXS7133	GATA31E08	
	Green:	GATA172D05		DXS7423	DXS6809	
	Yellow:	DXS7132	DXS9902	DXS6789		
F	**ITALIAN 12PLEX ASSAY (TURRINA et al. 2007)**					
	Blue:	DXS7132	DXS8378	DXS6809		
	Green:	DXS7133	DXS6789	DXS7424		
	Yellow:	GATA172D05		HPRTB	DXS7423	DXS6807
	Red:	Amel	GATA31E08	DXS101		
G	**BRAZILIAN 11PLEX ASSAY (RIBEIRO RODRIGUES et al. 2008)**					
	Blue:	DXS7132	DXS7423	DXS7133	DXS10011	
	Green:	DXS7130	DXS6800	GATA31E08	HPRTB	
	Yellow:	DXS6789	DXS9898	DXS9895		

(Continued)

TABLE 15.4 Configuration of Several X-STR Assay and Kits Reported in the Literature Showing Dye Color Labels and Relative Sizes of PCR Products. The Arbitrary Assay/Kit Designation (A, B, C, etc.) Is Used in Table 15.3. (*Continued*)

AFDIL MINIX-STR ASSAYS (DIEGOLI & COBLE 2011)				
H1	*Blue:*	DXS6789		
	Green:	DXS7130	DXS9902	
	Yellow:	SRY	GATA31E08	DXS7424
	Red:	GATA165B12		DXS101
H2	*Blue:*	DXS6795	GATA172D05	DXS10147
	Green:	DXS8378	DXS7132	
	Yellow:	SRY	DXS6803	HPRTB
	Red:	DXS7423	DXS9902	

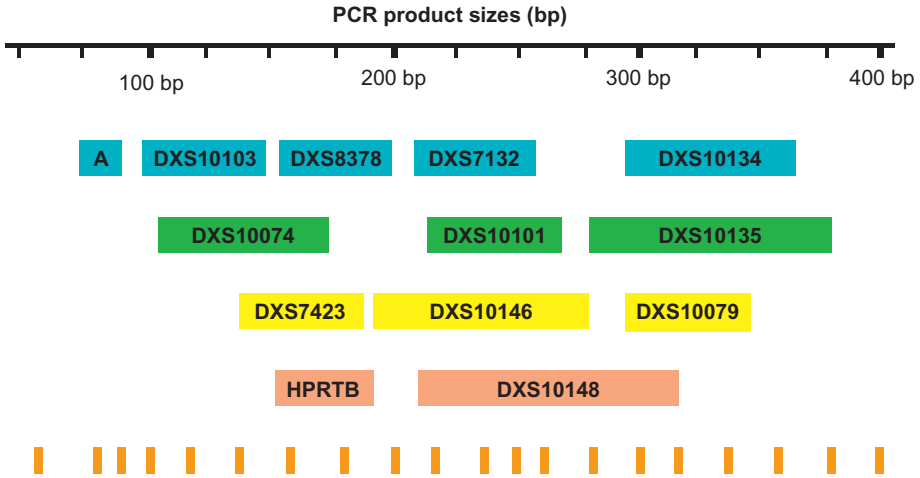


FIGURE 15.3 Investigator Argus X-12 kit (QIAGEN, Hilden, Germany) PCR product size ranges and dye color labels used for 12 X-STR loci and amelogenin (A).

Brazilian, Chinese, Finnish, German, Irish, Italian, Korean, Latin American, Latvian, Mongolian, Pakistani, Polish, Portuguese, Spanish, Taiwanese, Tibetan, UK, and U.S. populations.

Family inheritance studies have been performed (Hering et al. 2010, Tillmar et al. 2010) and even other species studied in terms of their ChrX variation (van Asch et al. 2010). While

X-chromosome analysis is still a niche application in forensic DNA typing, research efforts are ongoing to ensure its utility when it is needed.

READING LIST AND INTERNET RESOURCES

General Information

Forensic X-STR Research Website. <http://www.chrx-str.org/>.
Giltay, J. C., & Maiburg, M. C. (2010). Klinefelter syndrome: Clinical and molecular aspects. *Expert Review of Molecular Diagnostics*, *10*, 765–776.
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